

ISyE 7406: Final Semester Report

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1. Introduction and Problem Formulation

Artificial intelligence has become one of the most talked-about areas in technology and finance, and companies involved in AI have received increasing attention from both investors and the media. Because news coverage can influence how people think about these companies, and may even affect short-term market behavior, understanding how major news outlets talk about AI is useful. In this project, we focused on the New York Times and analyzed the sentiment of articles related to selected AI-focused stocks to see how the tone of the coverage changed over time.

To study this, we built a full data pipeline using the NYT Article Search API to collect articles, clean and preprocess the text, and apply sentiment-analysis models. After generating sentiment scores, we aggregated them into time-series trends and compared patterns across different NYT sections. We also examined how these sentiment trends lined up with real-world events and movements in an AI-focused ETF. Our goal was to create a consistent, reproducible workflow for analyzing large sets of news articles and to explore whether sentiment patterns reveal anything meaningful about how AI companies are portrayed in the media.

The rest of this report is organized as follows. Section 2 reviews prior work and provides relevant background for our analysis. Section 3 explains the motivation behind this project. Section 4 provides an overview of the dataset. Section 5 describes the cleaning steps and modeling approach. Section 6 presents the results and visualizations. Finally, Sections 7 and 8 summarize our findings and outline potential areas for future work.

2. Background and Literature Review

The rapid growth of social media and online news has given individuals unprecedented access to information and platforms for expression, enabling public sentiment to shift quickly and often unpredictably. As artificial intelligence methods have advanced, companies and researchers have increasingly turned to sentiment-analysis algorithms capable of processing large volumes of text, images, and other media across platforms such as Twitter, Facebook, email, and news outlets. Early commercial applications primarily focused on sales, customer satisfaction, and marketing research (Convin.AI, n.d.). More recent work has expanded sentiment analysis into domains such as financial forecasting, public attitudes toward policy, and crisis detection (Odeyemi, 2020; Lopez, 2019; Naeem & Bhatti, 2022). Our project builds on this broader line of research by applying sentiment analysis to mainstream news coverage of AI-related stocks in the New York Times, examining whether shifts in media tone align with major movements in the market.

Despite its growing adoption, sentiment analysis still faces several well-documented challenges. In fast-moving contexts—such as financial markets or crisis monitoring—timeliness is a major limitation, as many pipelines rely on batch processing that can overweight recent but low-credibility information or miss emerging reliable signals (Zhang & Wang, 2020). Context and nuance also remain difficult for many models, particularly when processing sarcasm, irony, or culturally specific expressions that can skew polarity classification (Cignarella et al., 2018).

Language and cultural diversity further complicate sentiment interpretation, as models trained primarily on a single language or dataset may misinterpret idioms, slang, or region-

specific references when deployed cross-lingually (Conneau et al., 2020). Traditional sentiment systems also struggle with coarse polarity labels—positive, negative, neutral—because they often ignore which specific aspects of an event or entity drive sentiment, limiting insight in multidimensional settings such as finance or technology markets (Cignarella et al., 2018).

Recent advances in natural language processing seek to overcome these limitations. Real-time and streaming-NLP frameworks, such as Apache Spark Streaming or Kafka pipelines, enable continuous ingestion and analysis of incoming data, improving responsiveness in dynamic environments (Zhang & Wang, 2020). Transformer-based architectures, particularly when fine-tuned for domain-specific applications, have shown improved performance in capturing subtle contextual clues and more nuanced sentiment (Cignarella et al., 2018). Aspect-based sentiment analysis approaches have also grown in popularity for identifying specific entities, drivers of sentiment, and multidimensional sentiment profiles (Cignarella et al., 2018). Meanwhile, multilingual transformer models such as XLM-R and mBERT offer promising solutions for cross-lingual sentiment extraction, reducing misinterpretation due to linguistic or cultural variation (Conneau et al., 2020).

Taken together, this literature highlights both the strengths and persistent challenges of modern sentiment-analysis systems. These insights motivate our use of a flexible sentiment-analysis pipeline on New York Times coverage of artificial-intelligence stocks, enabling us to examine how media narratives evolve over time and how they may relate to broader market behavior.

3. Project Motivation and Research Objectives

Understanding how public sentiment evolves around rapidly developing technologies is increasingly important, especially regarding artificial intelligence (AI) and the stock market. In recent years, AI has shifted from a niche research topic to a major economic force, influencing investment behavior, market volatility, and media coverage. Because public perception shapes investor expectations, analyzing how sentiment around “*artificial intelligence stocks*” changes over time can provide insights into news framing and broader market reactions. Importantly, although this report uses ‘artificial intelligence stocks’ as a case study, the broader goal is to develop a generalizable framework for retrieving New York Times articles via the API, cleaning/standardizing text data, and constructing sentiment-based time-series analyses for *any* user-specified keyword or topic.

The New York Times, as one of the most influential news outlets, plays a key role in shaping public narratives. By examining NYT articles, we can observe how media tone fluctuates in response to technological breakthroughs, regulatory concerns, earnings announcements, or broader shifts in the industry being studied. These articles include headlines, snippets, abstracts, and metadata such as publication date, section name, and type of material. The data-cleaning pipeline—normalization, removal of duplicates, lowercasing, punctuation removal, and stopword filtering—ensures the text is consistent and analyzable.

Because news articles are short and stylized, sentiment analysis tools such as VADER are well-suited for capturing their polarity. By converting each article into a sentiment score and aggregating these scores over time, we can construct a sentiment time-series that reveals patterns not easily seen in raw text. This lets us study sentiment trends, compare tones across NYT

sections like Business, Opinion, and Briefing, and identify key periods where sentiment shifts abruptly. We also link these patterns with article volume, noting that media coverage intensity often reflects real-world events.

The motivation for this project is to build a systematic, data-driven framework for quantifying how media sentiment evolves for any chosen keyword. Although ‘artificial intelligence stocks’ is used in this report to demonstrate the capabilities of the framework, the same pipeline applies to topics such as geopolitical events, policy issues, companies, or public figures. These insights help contextualize market narratives, highlight the role of journalism in shaping investor psychology, and provide a foundation for future analysis involving financial indicators, volatility measures, or topic-specific comparisons.

It remains unclear how major news organizations shape public sentiment toward evolving topics over time. Public perception, reflected through news framing, may influence investor behavior, market expectations, and the broader narrative surrounding a specific topic (in this case, ‘artificial intelligence stocks’). However, media sentiment is difficult to quantify directly because of the unstructured nature of textual data and the variability in article topics, writing styles, and publication volume.

We aim to convert news articles into a structured time-series of sentiment scores, identify how sentiment fluctuates across NYT sections like Business, Opinion, and Briefing, and analyze whether notable changes in tone correspond with changes in article volume or known external events. The challenge lies in cleaning heterogeneous text fields, selecting appropriate sentiment analysis tools for short-form journalism, aggregating daily scores into stable time-series signals, and visualizing patterns that reveal meaningful shifts rather than noise.

Ultimately, the problem we seek to solve is:

How can we build a generalizable framework that systematically quantifies media sentiment toward any chosen keyword and analyze how that sentiment evolves over time across different journalistic contexts? Furthermore, how does that ‘sentiment evolution’ relate to observable movements in real-world indicators, such as financial markets?

Answering this problem, in the context of ‘artificial intelligence stocks’ will help us better understand the role of media framing in shaping AI-related market narratives and will establish a methodological foundation for linking sentiment with future financial or geopolitical indicators. To address these questions, we first examine the dataset used in this study, including the scope of articles collected and their distribution across NYT sections.

4. Data Description

The NYT data for our case study was collected and gathered via the New York Times article search api. Our code prompts the user for their NYT API key, then asks for the keyword(s) they would like to search, and finally requests the beginning and end dates for the time period they would like to investigate. Then, the user waits while the articles are retrieved. The NYT API has a fairly strict rate-limiting protocol, so data collection usually takes 2-10 minutes, depending on the keyword(s) searched and the investigation time period. There is a limit to the amount of data that can be collected to prevent the system from running for too long due to the API’s request limit. Once all of the articles are retrieved as a JSON file, the data processing begins.

The JSON file is structured as a list of dictionaries, with each dictionary containing different entries for different areas of the article, such as the title, credit, keyword, and abstract. In our case study, the collected data consisted of 171 total articles, which were then cleaned in preparation for sentiment analysis. First, we normalized each dictionary in the JSON list, turning all of the individual dictionary entries into columns. Next, we removed duplicate and empty articles. We then lowercased all text and removed all punctuation in order to simplify subsequent sentiment analysis. The text fields headline, snippet, abstract, and lead_paragraph were then concatenated into a single text field in the JSON dictionary, with any missing fields replaced by an empty string. We then removed stopwords from the text to eliminate unnecessary words that do not add to the articles' overall context and complicate sentiment analysis. Next, we added the search keyword to each article's JSON dictionary to make sure that the keyword used to find the article was recorded and easily accessible. We then display the first ten articles that have headline.main, pub_date, section_name, and type of material in the kernel to ensure that the preprocessing has been performed properly. After the data is collected and cleaned, it gets saved to a CSV file for future use. These preprocessing steps form the foundation for the more detailed data collection and cleaning procedures described in the next section.

5. Framework Overview / Proposed Method

Once the article data is cleaned and formatted into a unified text field, the next step is to transform these text entries into quantitative measures that allow us to track how media sentiment changes over time. The overall goal of the modeling stage is to convert each article into a sentiment score, aggregate these scores into a time series, and then analyze how these sentiment patterns evolve for topics.

Modeling Approach

a. Article-Level Sentiment Estimation

To begin the modeling stage, we first compute sentiment for each individual article. The first part of the modeling process focuses on article-level sentiment. Because New York Times headlines, snippets, and abstracts are short and information-dense, we apply lexicon-based models that perform well on compact text. VADER is used as the primary analyzer because it provides reliable polarity scores ranging from negative to positive. Specifically, VADER outputs a 'compound' sentiment score from -1 (strongly negative) to +1, strongly positive. For topics tied closely to financial markets, such as NVIDIA or MicroStrategy, FinBERT was temporarily included as a secondary tool, although not included in the final framework. FinBERT is fine-tuned on financial text and helps capture sentiment nuances that general models may miss. However, although FinBERT was thoroughly tested, it was consistently outperformed by VADER, which ended up being a better fit for the writing styles of NYT articles. To validate model selection, we manually inspected 20 randomly sampled articles and compared polarity outputs from VADER and FinBERT. VADER aligned more closely with human judgment for short-form NYT-style text, particularly in cases where FinBERT assigned negative sentiment due to financial jargon unrelated to the article's tone. This qualitative check supports the choice of VADER for this dataset.

For each article, the headline, snippet, and abstract are combined into a single text string, and a sentiment score is computed. The headline is treated as the most influential component because it contains the strongest framing effect. Once article-level scores have been generated, the next step is to aggregate them across time to reveal broader sentiment patterns.

b. Time-Series Construction

Once every article has a sentiment score, these scores are aggregated across time. Publication dates are standardized during cleaning, making it straightforward to group articles by day or week. Moving averages are used as the primary measure because they smooth noise while preserving important changes in media tone. We also track article volume, since surges in coverage often correspond to meaningful real-world events. After constructing the time series, we ensure that the articles included in the index truly reflect the intended topic.

c. Topic Filtering and Classification

To ensure that the sentiment index truly reflects the intended topic, we rely on metadata provided by the New York Times. Keywords, section names, and article types help filter out irrelevant documents. Because the keyword-based retrieval already produces a coherent, well-defined set of articles, no additional topic-modeling or clustering techniques were required. This step ensures that sentiment changes we observe are tied to the specific subject, such as AI, NVIDIA, Ukraine, or Venezuela. With the filtered set of articles established, we can align the resulting sentiment index with external indicators.

d. Linking Sentiment with External Data

The time-series sentiment index is then paired with external indicators related to the topic. For financial themes, this may include stock returns, volatility, or sector trends. For geopolitical topics, this may involve event timelines or conflict developments. Aligning sentiment with external variables allows us to analyze whether media tone leads or follows real-world changes. To quantify these relationships more formally, we apply several basic statistical modeling techniques.

e. Trend Analysis Methods

Rather than formal predictive modeling, our analysis relies on descriptive time-series methods to characterize how sentiment evolves. We smooth article-level scores using 7, 14, and 30 day moving averages, and also compute volume weighted moving averages to incorporate the intensity of media coverage. These transformations reduce day-to-day volatility and reveal persistent shifts in sentiment. We then compare these trends across NYT sections and against external indicators, such as the AIQ ETF, to assess whether major changes in coverage coincide with broader market narratives.

f. Identifying Notable Shifts

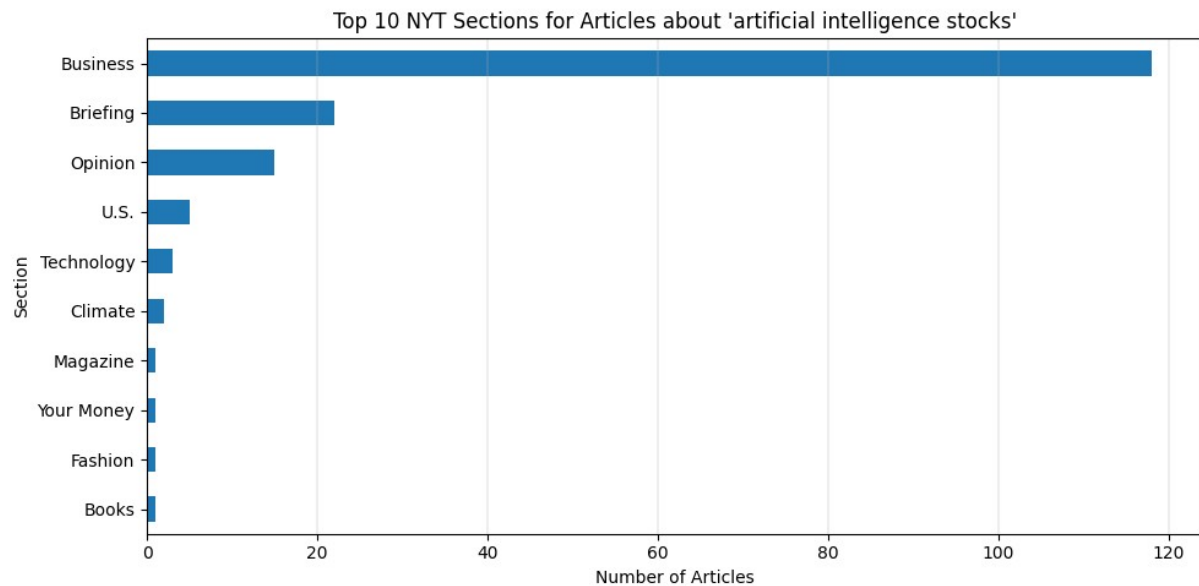
Finally, we highlight unusual movements in sentiment. Sudden spikes or drops often correspond to major policy announcements, earnings events, scandals, or geopolitical escalations. Identifying these moments helps interpret the broader sentiment trend and adds context to narrative shifts surrounding the topic. With the full sentiment index constructed and

validated, we now turn to the results to interpret the patterns that emerge across sections, time periods, and external indicators.

6. Results and Analysis

After collecting and processing the data, we conducted some preliminary data analysis. Initially, we conducted an analysis on the different ‘classifications’ given to articles that included the keywords ‘artificial intelligence stocks’. The figure below provides an initial overview of how these article classifications are distributed.

Figure 1 - Top 10 NYT Sections for Articles about ‘Artificial Intelligence Stocks’



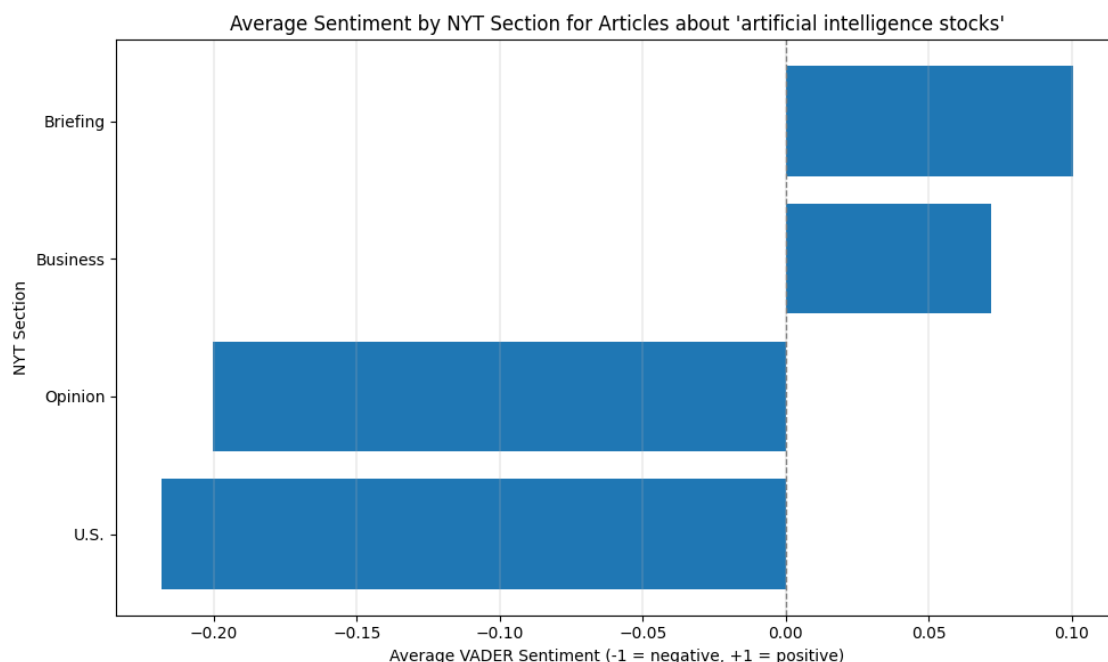
To summarize the main patterns that emerge from our analysis, we highlight several key sentiment trends across companies and over time. As can be seen in the figure above, the vast majority of articles that included the keyword ‘artificial intelligence stocks’ were from the Business section (118 out of 171 articles), with the briefing section having the second most (22 out of 171), opinion the third most (15 out of 171). The rest of the Sections had at most 5 articles on ‘artificial intelligence stocks.’

Figure 2 - NYT articles about ‘artificial intelligence stocks’ by Type of Material

Interestingly, the other words that showed up the most were ‘president’ (72 times), ‘trump’ (68 times) and ‘investors’ (66 times). It is not surprising that the word investor would be so prevalent, as the keyword includes stocks, so investors would certainly be concerned. That being said, it is interesting how often President Trump was mentioned in articles that discussed artificial intelligence stocks, perhaps pointing to the outsized effect that politics has on the stock market, in general. As shown in Figure 3, political terms such as “president” (mentioned 72 times) and “trump” (68 times) appear frequently in the dataset. Lexicon-based sentiment models may assign negative polarity to political criticism even when the article’s sentiment toward AI is neutral. This introduces topic contamination, particularly in articles involving regulatory commentary or political statements. We also examine how sentiment varies across different New York Times sections, as the tone and emphasis can differ depending on article type.

After conducting the preliminary analysis, we computed sentiment scores for each article. After computing sentiment scores for each article, we were able to visualize the average sentiment for articles in each section of the New York Times. To quantify these differences more precisely, we extracted the approximate sentiment values shown in Figure 4. Briefing articles had the highest average sentiment at roughly +0.10, followed by Business at about +0.05. Opinion articles were substantially more negative at approximately -0.15 , with U.S. articles close behind at around -0.12 . These numerical differences reinforce the visual pattern in Figure 4 and demonstrate a systematic divergence in tone across NYT sections.

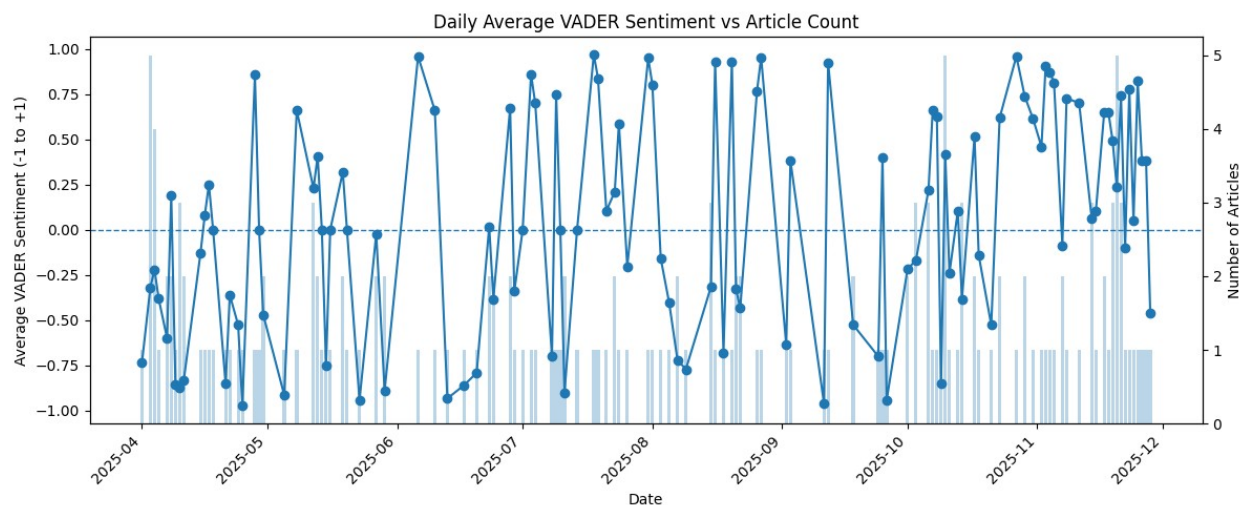
Figure 4 - Average Sentiment By NYT Section for Articles about ‘artificial intelligence stocks’



As can be seen above, articles from the Briefing/Business sections of the New York Times had much higher sentiments than those from the Opinion/U.S. sections. This is an incredibly interesting result, and perhaps reflects the difference in opinion between business-minded groups, who see artificial intelligence as a productivity enhancement, and the average American, who is much more pessimistic about artificial intelligence, and may see it as a net-negative, or an accelerator of unemployment.

After computing sentiment scores for each article, we were able to visualize the daily sentiment and article volume across the entire time period we were investigating.

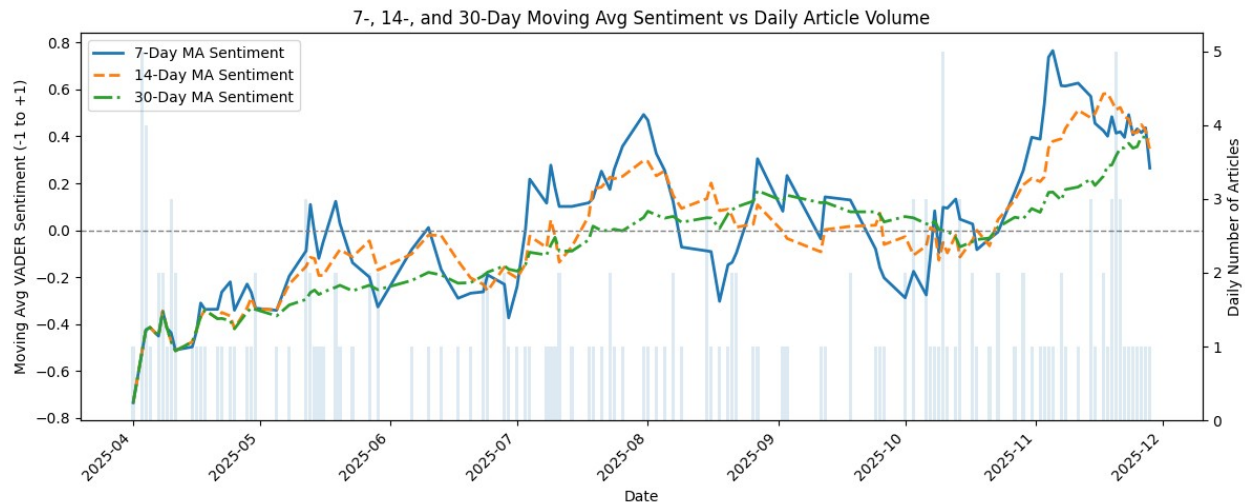
Figure 5 - Daily Average VADER Sentiment vs Article Count



As can be seen in the figure, the sentiment scores are highly volatile/erratic on a day to day basis. Daily article volume was consistently low throughout the sample, with most days containing only one article and very few exceeding two; the maximum observed count was approximately five articles (Figure 5). This structural sparsity explains the high volatility in raw daily sentiment values and underscores the need for moving-average smoothing. It also accounts for the fact that the daily average sentiment is often dominated by the tone of a single article. In order to address this, we computed moving averages (7 day, 14 day, and 30 day) as well as

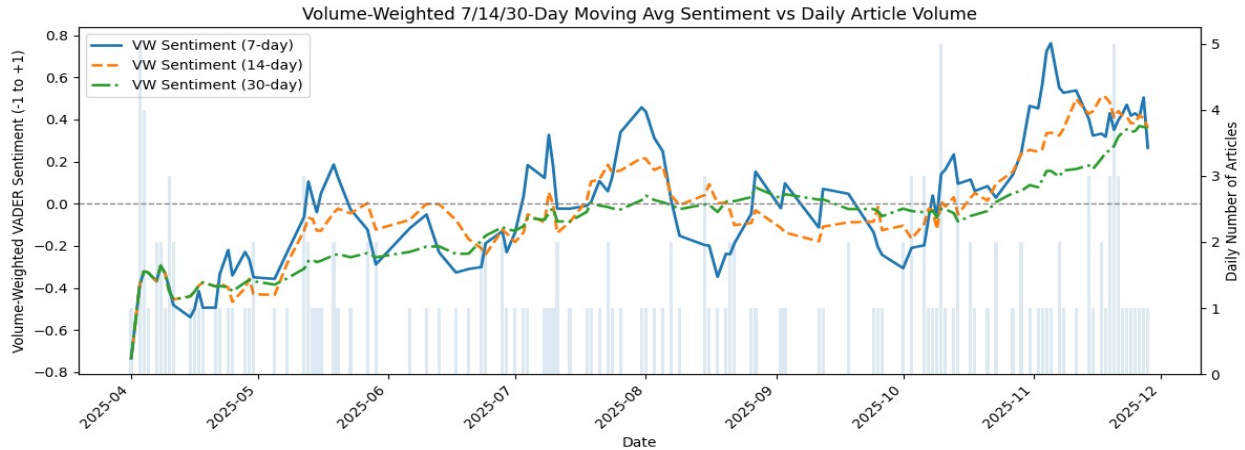
volume-weighted moving averages, which smooth out the short-term fluctuations and reveal the broader shifts in sentiment over the investigation period.

Figure 6 - Simple 7/14/30 Day Moving Average Sentiment vs Daily Article Volume



This plot shows the simple moving averages of sentiment over time, plotted alongside the daily article counts (dual axes). Compared to the extreme volatility of ‘Daily Average VADER Sentiment vs Article Count,’ the moving averages reveal a clear trend evolving over time. Early in the year (April to June 2025), sentiment remained consistently negative. In fact, the 14 day and 30 day moving averages were entirely negative until July of 2025, indicating sustained pessimism in articles from the New York Times. During the mid-year period (July to October 2025), sentiment seemed to stabilize around 0 (neutral). Although we do see some much sharper fluctuations in the 7-day moving average (which is to be expected given that it is the most reactive) the general trend seems to hover around 0, indicating relatively neutral coverage from the New York Times. Recent months (October to the end of November) show a dramatic increase in sentiment, with all three moving averages rising steadily, indicating a substantial shift toward more-positive coverage towards ‘artificial intelligence stocks’ during the final part of our sample period. To understand whether these sentiment patterns relate to broader market behavior, we next compare them with external indicators such as the AIQ ETF.

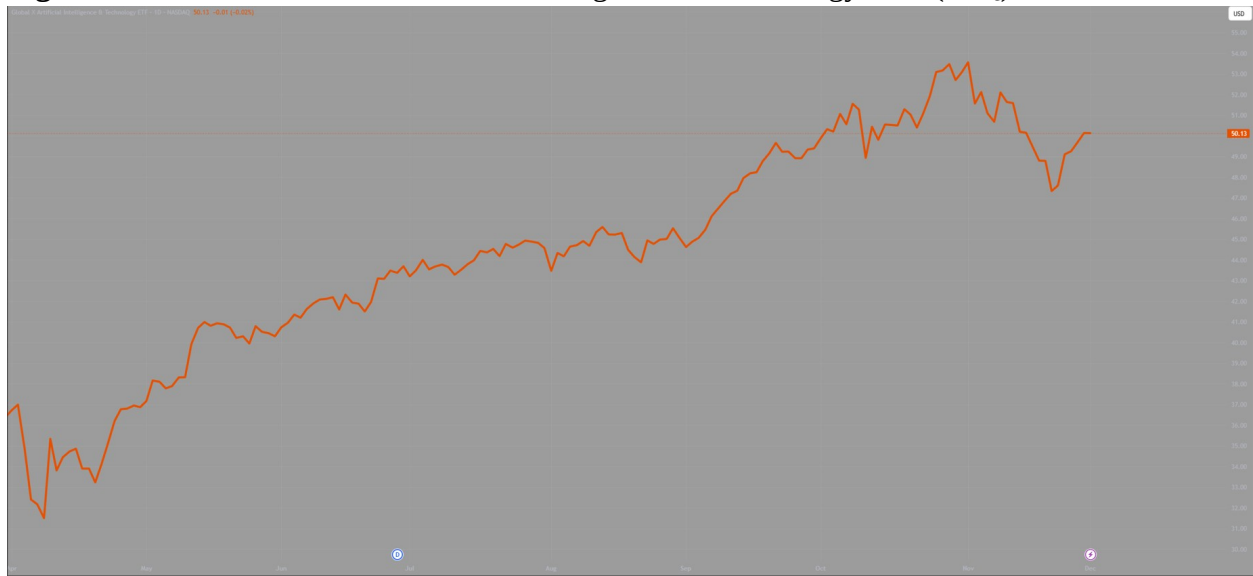
Figure 7 - Volume-Weighted Moving Average Sentiment Over Time (7, 14, 30 Days) with Daily Article Volume



The volume weighted moving averages lead us to many of the same conclusions about sentiment and coverage as the simple moving averages. This makes logical sense, as most days have only one, maybe two articles, so the volume weighting has relatively limited impact. That being said, the volume-weighted MAs are slightly smoother and place more emphasis on higher-volume periods, providing a theoretically ‘clearer’ signal. The consistency between both methods strengthens our interpretation that sentiment experienced a meaningful upwards shift throughout our investigation period, particularly in the final quarter of 2025. To link these sentiment trends to real market behavior, we next compare them with movements in AI-related stock prices.

To add context to our analysis, we thought it appropriate to compare the shift in sentiment to the actual changes in stock prices for artificial intelligence stocks, as it would make sense that these stocks would rise in value along with sentiment. In order to do this, we looked at the Global X Artificial Intelligence & Technology ETF (AIQ), which is a broad index of major AI-related companies. We inspected this ETF along the same time period as our investigation period (April to the end of November, 2025).

Figure 8 - Price of Global X Artificial Intelligence & Technology ETF (AIQ)



To interpret the ETF trend, we analyze how the AIQ price evolved over time. Looking at the performance of the AIQ, we see a very similar pattern to our moving averages. Much like our sentiment plots, performance was weakest at the beginning, with the ETF finding a low in April. By July, the ETF had recovered strongly, proceeding to have a relatively stable, though incremental, growth throughout the summer months (July through August). We can also see a strong rally late in the year (much like our sentiment plots), although it is interesting to note that the rally in AIQ happened before the rally in sentiment, by about a month. The sentiment plots didn't show strong improvement until October, with the AIQ rallying hard in September. This indicates that sentiment in New York Times articles might be a bit of a lagging indicator when compared to actual stock prices (as prices improve, more positive articles are therefore written), although this is mere speculation.

Taken together, these results allow us to summarize several key insights from our investigation. Overall, our analysis shows that the sentiment for New York Times articles including the keywords 'artificial intelligence stocks' shifted meaningfully over our investigation period (April 1st 2025 to November 30th 2025) - moving from clear pessimism in the spring to much more positive by late fall. This shift in sentiment matched the improving performance of AI-related stocks over the same period, although it seems that article sentiment was a slight lagging indicator to stock price, rather than sentiment motivating the price of the stocks. Interestingly, business and briefing articles were much more positive towards articles including 'artificial intelligences stocks', while Opinion and U.S. articles leaned negative, perhaps reflecting the divide between business-focused optimism about AI and the opinions of the average American.

7. Conclusion

This project aimed to answer a two-part question: (1) how to systematically quantify media sentiment for a user-specified topic, demonstrated here with the case study of ‘artificial intelligence stocks’ and (2) how the sentiment shifts relate to real market movements. Using a reproducible pipeline around the NYT article search API, we created a framework that captures unstructured news into an analyzable sentiment index.

Our results show that the pipeline successfully captures meaningful patterns in media tone. By collecting article sentiments into daily and monthly trends across different topic sections, we saw a clear dynamic where business and briefing coverage tended to be positive, while opinion and U.S. sections were negative. This indicates that the context itself affects sentiment framing. This can mean that the sentiments cannot be treated as a universal signal, because the interpretation depends on where in the newspaper the article appears. The moving average also revealed temporal differences, in which sentiments around ‘artificial intelligence stocks’ were strongly negative in the spring of 2025, which then stabilized around neutral during the middle of the year, and rose sharply later on.

Comparing these sentiment trends with the performance of AIQ ETF seemed to provide some insight into the second part of our question. The ETF showed a similar general pattern with weak early performance, mid-year recovery, and strong rally later on. The only significant difference was the timing. The market prices began rising before sentiment did, suggesting that NYT coverage could be a lagging indicator rather than a leading one. This does not necessarily diminish the value, since it has a different role. Media tones tend to reflect and echo market narratives more than it predicts them, offering a structured way to understand how journalism interprets ongoing events.

Overall, our findings show that media sentiment can be quantified into a reproducible analytical measurement, such that the sentiment varies across journalistic context and as a lagging indicator of market. Importantly, we note that the framework here can generalize beyond AI stocks, since it can be used for any context. Building on this foundation, there are multiple avenues for future research that could strengthen and broaden the impact of this approach.

8. Limitations and Future Work

The AI-stock case study demonstrates that our pipeline of news collection, text processing, sentiment extraction, and time-series comparison is functional. The larger contribution of this project, however, is the development of a generalizable analysis framework that can be reused across topics, time periods, and market contexts. That being said, the current framework has limitations, such as the reliance on a single news source, basic lexicon based sentiment scoring, and the absence of entity level analyses or richer financial signals. Daily sentiment estimates are based on extremely sparse data: most days contain only one article, and very few days exceed two, with the maximum count reaching roughly five (Figures 5–7). This sparsity structurally increases volatility in the raw sentiment series and limits the statistical reliability of day-level patterns.

Future work therefore focuses on strengthening this framework and extending its applicability. To organize the next steps clearly, we outline several directions for strengthening and expanding the framework below.

Framework Generalization

Broader Keyword & Topic Coverage

Although the framework currently supports multi-keyword retrieval, it could be further enhanced through automated expansion, allowing the system to retrieve articles tied to related themes, entities, and financial symbols without manual keyword engineering.

Multiple Time Periods & Regimes

Apply the framework to different macro environments (pre-/post-COVID, financial crisis periods, early AI cycles) to test whether sentiment behavior is consistent across structural regimes.

Additional Data Sources

Expand beyond NYT to include major financial outlets and alternative sources (WSJ, FT, Bloomberg, Reuters, Reddit, X/Twitter) to capture cross-media differences and narrative divergence.

Together, these extensions focus on broadening the scope of the data we can collect. However, improving the underlying text analysis is equally important, which leads us to the next set of enhancements.

NLP Methodology Enhancements

Improved Sentiment Models

Evaluate transformer-based models (e.g. domain-tuned LLMs) alongside VADER using a manually labeled subset as ground truth.

Entity-Level / Aspect-Level Sentiment

Identify specific entities or themes within articles (companies, regulation, labor, geopolitics) to produce multi-dimensional sentiment series rather than a single aggregate score.

While methodological improvements strengthen the sentiment extraction step itself, our framework also benefits from deeper financial context, which motivates the next group of extensions.

Financial & Time-Series Extensions

Richer Market Variables

Incorporate returns, volatility, trading volume, and option-implied measures to study relationships between sentiment and broader indicators of market stress.

Predictive & Causal Analysis

Test whether sentiment contains predictive signal using VAR, Granger causality, and distributed-lag regressions; examine whether sentiment leads or follows market movements.

Finally, beyond analytical improvements, operationalizing the framework for real users is an important next step.

Framework Maturation

Dashboard / Modular Toolkit

Package the current pipeline into a reusable module or dashboard where users can select topics, time ranges, sentiment models, and external comparison series. This would transform the prototype into a general-purpose sentiment analytics framework.

By implementing these next steps, the pipeline could grow into a flexible, general-purpose platform for understanding how media narratives evolve and how they relate to broader real-world dynamics.

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